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26TH.py - C:/Users/thala/Downloads/FODS/26TH.py (3.14.3)
File Edit Format Run Options Window Help
from sklearn.linear_model import LinearRegression

# Training data: [Area, Bedrooms]
X = [
    [1000, 2],
    [1200, 2],
    [1500, 3],
    [1800, 3],
    [2000, 4]
]

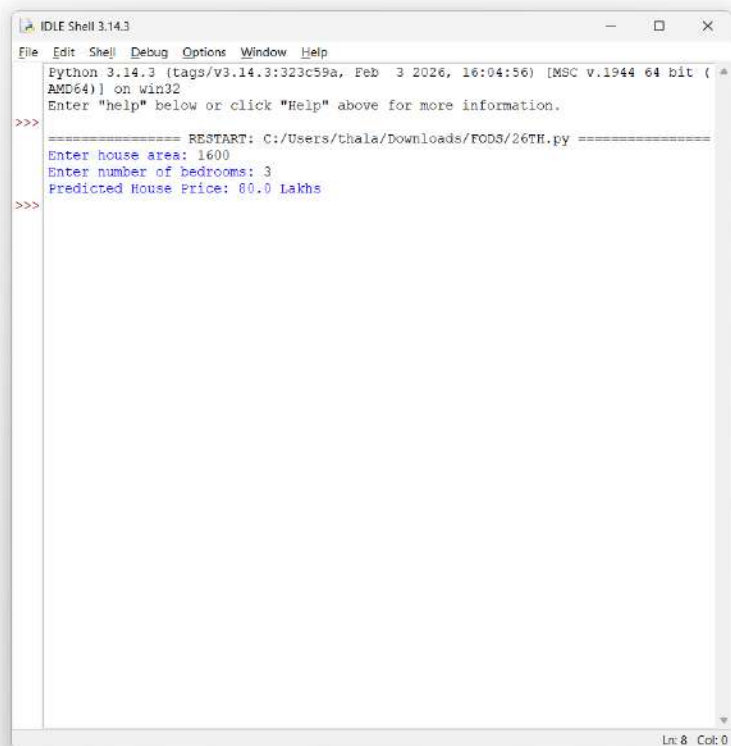
# House prices
y = [50, 60, 75, 90, 100]

# Train model
model = LinearRegression()
model.fit(X, y)

# User input
area = float(input("Enter house area: "))
bedrooms = int(input("Enter number of bedrooms: "))

# Prediction
price = model.predict([[area, bedrooms]])

print("Predicted House Price:", round(price[0], 2), "Lakhs")
```



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IDLE Shell 3.14.3
File Edit Shell Debug Options Window Help
Python 3.14.3 (tags/v3.14.3:323c59a, Feb  3 2026, 16:04:56) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>>
===== RESTART: C:/Users/thala/Downloads/FODS/26TH.py =====
Enter house area: 1600
Enter number of bedrooms: 3
Predicted House Price: 80.0 Lakhs
>>>
```

```

27TH.py - C:/Users/thala/Downloads/FODS/27TH.py (3.14.3)
File Edit Format Run Options Window Help
from sklearn.linear_model import LogisticRegression

# Training data: [Usage Minutes, Contract Duration]
X = [
    [100, 24],
    [120, 18],
    [150, 12],
    [200, 6],
    [250, 3],
    [300, 2]
]

# 0 = No Churn, 1 = Churn
y = [0, 0, 0, 1, 1, 1]

# Train model
model = LogisticRegression()
model.fit(X, y)

# User input
minutes = float(input("Enter usage minutes: "))
contract = float(input("Enter contract duration: "))

# Prediction
prediction = model.predict([[minutes, contract]])

if prediction[0] == 1:
    print("Customer Will Churn")
else:
    print("Customer will not Churn")

```

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>>>
===== RESTART: C:/Users/thala/Downloads/FODS/26TH.py =====
Enter house area: 1600
Enter number of bedrooms: 3
Predicted House Price: 80.0 Lakhs

>>>
===== RESTART: C:/Users/thala/Downloads/FODS/27TH.py =====
Enter usage minutes: 250
Enter contract duration: 3
Customer will Churn

>>>

```

Ln 13 Col 0

```

28TH.py - C:/Users/thala/Downloads/FODS/28TH.py (3.14.3)
File Edit Format Run Options Window Help
from sklearn.cluster import KMeans

# Customer data: [Amount Spent, Number of Purchases]
X = [
    [1000, 5],
    [1200, 6],
    [900, 4],
    [5000, 20],
    [5500, 22],
    [6000, 25]
]

# Create model
model = KMeans(n_clusters=2, random_state=42, n_init=10)
model.fit(X)

# User input
amount = float(input("Enter amount spent: "))
purchases = int(input("Enter number of purchases: "))

# Predict cluster
cluster = model.predict([[amount, purchases]])

print("Customer belongs to Segment:", cluster[0])

```

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>>>
===== RESTART: C:/Users/thala/Downloads/FODS/26TH.py =====
Enter house area: 1600
Enter number of bedrooms: 3
Predicted House Price: 80.0 Lakhs
>>>
===== RESTART: C:/Users/thala/Downloads/FODS/27TH.py =====
Enter usage minutes: 250
Enter contract duration: 3
Customer will Churn
>>>
===== RESTART: C:/Users/thala/Downloads/FODS/28TH.py =====
Enter amount spent: 5200
Enter number of purchases: 21
Customer belongs to Segment: 1
>>>

```

29TH.py - C:/Users/thala/Downloads/FODS/29TH.py (3.14.3)

File Edit Format Run Options Window Help

```
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score
```

```
# Load dataset
data = load_iris()
```

```
X = data.data
y = data.target
```

```
# Split data
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

```
# Train model
model = DecisionTreeClassifier()
model.fit(X_train, y_train)
```

```
# Predictions
y_pred = model.predict(X_test)
```

```
# Evaluation
print("Accuracy:", accuracy_score(y_test, y_pred))
print("Precision:", precision_score(y_test, y_pred, average="weighted"))
print("Recall:", recall_score(y_test, y_pred, average="weighted"))
print("F1-Score:", f1_score(y_test, y_pred, average="weighted"))
```

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```
>>> ===== RESTART: C:/Users/thala/Downloads/FODS/26TH.py =====
Enter house area: 1600
Enter number of bedrooms: 3
Predicted House Price: 80.0 Lakhs
```

```
>>> ===== RESTART: C:/Users/thala/Downloads/FODS/27TH.py =====
Enter usage minutes: 250
Enter contract duration: 3
Customer will Churn
```

```
>>> ===== RESTART: C:/Users/thala/Downloads/FODS/28TH.py =====
Enter amount spent: 5200
Enter number of purchases: 21
Customer belongs to Segment: 1
```

```
>>> ===== RESTART: C:/Users/thala/Downloads/FODS/29TH.py =====
Accuracy: 1.0
Precision: 1.0
Recall: 1.0
F1-Score: 1.0
>>> |
```

Ln: 24 Col: 0

```

30TH.py - C:/Users/thala/Downloads/FODS/30TH.py (3.14.3)
File Edit Format Run Options Window Help
from sklearn.tree import DecisionTreeRegressor

# Training data: [Mileage, Age]
X = [
    [20000, 2],
    [30000, 3],
    [40000, 4],
    [50000, 5],
    [60000, 6]
]

# Car prices
y = [800000, 700000, 600000, 500000, 400000]

# Train CART model
model = DecisionTreeRegressor(random_state=42)
model.fit(X, y)

# User input
mileage = float(input("Enter car mileage: "))
age = float(input("Enter car age: "))

# Predict price
price = model.predict([[mileage, age]])
print("Predicted Car Price:", price[0])

# Display decision path
path = model.decision_path([[mileage, age]])
print("Decision Path Nodes:", path.indices)

```

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AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/thala/Downloads/FODS/26TH.py =====
Enter house area: 1600
Enter number of bedrooms: 3
Predicted House Price: 80.0 Lakhs
>>>
===== RESTART: C:/Users/thala/Downloads/FODS/27TH.py =====
Enter usage minutes: 250
Enter contract duration: 3
Customer will Churn
>>>
===== RESTART: C:/Users/thala/Downloads/FODS/28TH.py =====
Enter amount spent: 5200
Enter number of purchases: 21
Customer belongs to Segment: 1
>>>
===== RESTART: C:/Users/thala/Downloads/FODS/29TH.py =====
Accuracy: 1.0
Precision: 1.0
Recall: 1.0
F1-Score: 1.0
>>>
===== RESTART: C:/Users/thala/Downloads/FODS/30TH.py =====
Enter car mileage: 30000
Enter car age: 3
Predicted Car Price: 700000.0
Decision Path Nodes: [0 1 3]
>>>
Ln: 30 Col: 0

```